



OPEN Diagnosis and classification of neuromuscular disorders using Bi-LSTM optimized with grey Wolf optimizer for EMG signals

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Hand recognition, the process of identifying or characterizing human hands in images or video streams, plays significant role in the biometrics, robotics, computer vision, and human-computer interaction. This technology relies on analyzing hand attributes such as shape, size, color, texture, and motion to perform tasks as gesture identification, hand tracking, and sign language interpretation. In particular, hand movement decoding from electromyography (EMG) signals has shown promise for understanding neuromuscular function and aiding in diagnosis and therapy for neuromuscular issues. Existing approaches range from deep learning techniques such as Long Short-Term Memory (LSTM) and Bidirectional LSTM (Bi-LSTM) to conventional machine learning methods like Support Vector Machines (SVM) and Random Forest. Deep learning automates the process, reducing the dependency on manual feature extraction. However, the performance of these models is heavily influenced by hyperparameters such as the number of neurons, hidden layers, and learning rates. This study proposes a novel method that uses the Grey Wolf Optimizer (GWO) to fine-tune the hyperparameters of a Bi-LSTM-based EMG classification system. Implemented in MATLAB R2021a, this approach aims to enhance the accuracy of Bi-LSTM models in categorizing EMG signals. Performance metrics such as accuracy of 95%, precision of 93%, F1-score of 94%, and recall of 91% are used for thorough evaluation. By leveraging GWO for hyperparameter optimization, the study aims to achieve more accurate diagnosis and efficient tracking of rehabilitation outcomes for patients with neuromuscular disorders. This research demonstrates the potential of integrating biomedical engineering and computational intelligence to empower individuals with neuromuscular disabilities, thereby enhancing their quality of life.

Keywords Hand recognition, EMG signals, Bi-LSTM, Grey Wolf optimizer, Deep learning, Neuromuscular disorders, Rehabilitation, Biomedical engineering, Assistive technology, Computational intelligence

The human hand has a truly wonderful multipurpose instrument that is essential for daily living. The hand is very important in all aspects of human life, and it is unparalleled in the entire animal kingdom in its multidimensionality, dexterity, and social communicative capabilities. Further, the hand is an effective word for social communication. Express feelings, send messages, and establish relationships with humans' help from nonverbal clues analogy gestures and handshakes. The hand is a very important instrument of interpersonal communication and expression due to its versatile complex movements and gestures that are capable of conveying nuance of meaning and emotion that words cannot¹. The hand recognition identifies or describes human hands in images or videos and is important for various applications like biometrics, robotics, computer vision, and human-computer interaction. Electromyography is the technique through which the electrical activity of the skeletal muscles is measured and recorded. Electrodes are applied to the skin above the targeted muscles in order to capture electrical impulses generated by muscle fibers during contraction and relaxation. These signals provide crucial data regarding biomechanics of human movement and also timing, patterns, and intensities of muscle activations².

The use of EMG sensors in prosthetic devices enables users to control their prostheses with signals from those muscles remaining in their limbs. Besides that, EMG pertains to the diagnostic use of determining the status

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of muscle function to find disorders that are related to neuromuscular control. The conditions in which much weakness is found or nerve damages as well as abnormality approaches in the movement can be known through EMG recordings. Such a diagnosis can give way to treatment patterns and rehabilitation strategies³. Detection of human gestures that can be further extended to hand tracking and even sign language interpretation is possible with the well-defined identification and analysis of features associated with hands, such as shape, size, color, texture, and movement. In computer vision, hand recognition systems employ the methodology of identifying and classifying hands in images or video streams by image segmentation, feature extraction, pattern recognition, and machine learning. The CNS motor neurons produce myoelectric signals, which are effective indicators of the neuromuscular function and are important in diagnosing several neurological and muscular diseases⁴.

EMG signals serve as important indicators in measure of the progress made through rehabilitative therapy. Several past techniques involved deep learning methods including Long Short-Term Memory (LSTM), Bidirectional LSTM, (Bi-LSTM) and hybrid models for decoding hand movements from EMG signals. Unlike normal machine learning processes that solely depend on feature extraction and selection, deep learning processes excelled on an automated feature learning aspect. Hyperparameters such as number of neurons, hidden layers, and learning parameters have a great impact on the performance of deep learning models. Optimum performance can only be achieved on the optimization of certain hyperparameters⁵. The work in this study presents a novel application of the Grey Wolf Optimizer (GWO) to optimize hyperparameters for a Bi-LSTM based EMG classification system. The aim of this optimization process is to improve classification accuracy of Bi-LSTM based EMG signals which will, in turn, produce accurate diagnosis and improved monitoring of rehabilitation outcomes in patients with neuromuscular diseases. Upgrading the performance and reliability of EMG-based hand recognition systems is a major advance in the field of augmentative communication and mental health habilitation technologies.

Previous work

Karnam et al. (2022)⁶ suggests a hybrid architecture called as dubbed EMGHandNet for accurately identifying the hand activities by utilizing the surface Electromyography (sEMG) signals. It integrates the bidirectional long short-term memory (Bi-LSTM) and CNNs. The Bi-LSTM encodes the sequential data in both forward and backwards to extract the temporal data and the CNN layers extract the spatial properties from the input signals. The end-to-end model significantly learns the inter-channel and bidirectional temporal data. The results of this model are compared with the existing models based on five datasets and ensures a better classification accuracy.

Rani et al. (2023)⁷. This study significantly underlines the necessity of EMG data in detecting the human activity (HAR) based on the several biological and therapeutic environments. The study evaluates the evolution of EMG-based activity detection and the integration of Artificial Intelligence (AI) method by clearly choosing the diverse set of the research papers. It analyses the feature extraction method, that more crucial for defining the muscle activity patterns, and data is pre-processed using advanced methods. The paper effectively identifies the HA from the EMG signals with the help of ML and DL models.

Ozdemir et al. (2022)⁸. The study compares transfer learning and time-frequency (TF) images of spontaneous surface electromyography (sEMG) signals to classify hand gestures. The 4-channel sEMG data were collected from 30 subjects as they performed seven different hand motions. TF pictures are extracted using three TF analysis techniques: STFT, CWT, and HHT. These images utilised in the combination with some pre-trained CNN model for accurate feature extraction. The cross-validation method is utilized to assess the results.

Erazo et al. (2023)⁹ Offers an long short-term memory networks (LSTM) model for effectively classifying the six grab kinds. Unlike the traditional model, this method serves an end-to-end deep learning results, removes the unwanted feature from the signals. The proposed LSTM-based method is surpassing other existing machine learning models on similar dataset based on the classification accuracy metrics. The results showcase the better performance in terms of higher F1-score with a significant result by minimizing the overfitting. The study ensures the significant efficacy of DL algorithms in enhancing the grasp type recognition from sEMG signals.

Lu et al. (2021)¹⁰. This study addresses the human lower limb leap phases identification from the surface electromyography (sEMG) signal for operating exoskeleton devices and evaluating the rehabilitation surfaces. The ConvLSTM model offers a significant online model by precisely captures the advanced spatiotemporal characteristics by the coupling of PCC matrix sequence with the sensor data confusion as input. The research introduces the synthesis method for Euler angle signals to enhance the recognition of human movement intention, and the investigation should impact the analysis of window number and length on model performance. Overall, this study offers a significant way for utilizing the LSTM networks to reliably and precisely identifies the lower limb leap phases from the sEMG signals. This method is more useful for exoskeleton control systems and rehabilitation purposes.

Li et al. (2023)¹¹. This study introduces the Internet of Medical Things (IoMT) and blockchain technology for enhancing the interoperability, user security and convenience. To overcome the limitations of the process in gesture detection, this study introduces the advanced method using dynamic EMG. The study suggests the Gaussian mixture model and upgraded with comb filter to significantly minimises the signal complexity in surface of EMG signals. Then, the extended regression model and temporal networks utilized for improving the continuous dynamic gesture recognition methods. The models combine seamlessly with the block chain-enabled IoMT systems, improving the data security and guaranteeing the data sharing traceability. The method enables a precise continuous dynamic gesture recognition while effectively reducing the false recognition rates. Overall, the method ensures a result for accurate and safe continuous dynamic gesture detection in the medical and the healthcare applications for deployment of block chain-enabled IoMT systems.

Ma et al. (2024)¹². The study suggests the novel feature fusion network (FFCSLT) using the sEMG data to precisely recognizing the traffic police gestures. The 6-channel sEMG sensor is employed for collecting the signals

during human movement. The dataset containing 20,000 sets of hand gesture data specific to traffic police (TPG). To enhance spatial features, the squeeze-and-excitation block (SE) is employed and the depth-wise separable convolutional network (DSCN) utilized for FFCSLT network. Furthermore, the temporal data is extracted by utilizing the LSTM-integrated Temporal Convolutional Network (TCN). The proposed method significantly ensures the proven by number of ablation study, that yields a better accuracy and reliable performance, ensures a reliable utilisation of sEMG-based gesture identification, in traffic enforcement contexts.

Raurale et al. (2020)¹³. This study ensures a difficulty in wrist-hand movements detection from the EMG data collected at the forearm's skin surface, while the sensors randomly deployed. It shows that state-of-the-art performance in wrist-hand pose detection may be accomplished with only 10% of the computing complexity required by standard systems, despite random sensor placement. Additionally, by utilizing low-cost embedded processors on wearable EMG devices, like the ARM Cortex-A53 processor, the suggested method permits real-time detection while cutting execution time by 90% without sacrificing detection performance. The research also investigates ways to further minimize computational burden by 30% by using fewer EMG channels or characteristics without sacrificing reliable identification capabilities. As the first example of real-time, high-performance wrist-hand movement detection for an independently battery-powered EMG wearable, this study represents a significant leap.

Jaramillo et al. (2020)¹⁴. The state-of-the-art in real-time hand gesture detection models using machine learning and electromyography (EMG) data is examined in this comprehensive overview of the literature. The Kitchenham is employed for selecting and evaluating the 65 major papers. The study offers an insightful data from the current status of EMG-based gesture recognition model by deconstructing the structure of advance method and enhancing the concepts in several domains, as data collection, pre-processing, feature extraction, classification, and the metrics analysis. The paper precisely detects the patterns and knowledge gaps in a gesture identification utilising the EMG data, offering a potential direction for the further research and enhancements in the HCI (human-computer interaction) technology.

Lu et al. (2020)¹⁵. This study utilizes electromyography (EMG) signals for personal recognition to addresses the growing necessity of insightful information security in digital age. The traditional method utilizes PINs and ID tags to hacking, biometric technologies that depends on the individual physiological attributes for the precise authentication and more widespread utilisation. EMG signals show potential for aliveness identification and thwarting spoofing assaults because of their link with living bodily features, even though widely used biometrics like voice recognition, iris scans, fingerprints, and facial recognition can still be fabricated. The report looks into both EMG-based personal identification and verification techniques, despite the paucity of research in this field. Myo armbands are applied to the right forearms of 21 participants in the study in order to record surface EMG signals when the hands are open. For EMG-based personal identification, two different approaches are suggested: one that makes use of the Discrete Wavelet Transform (DWT) and the ExtraTreesClassifier, and the other that makes use of the Continuous Wavelet Transform (CWT) and the Convolutional Neural Networks (CNN).

Puruhothaman et al. (2018)¹⁶. This study utilized for accurately recognizing the dexterous control of prosthetic hands from the EMG data. Initially, the EMG data is analyzed using the time domain features extracted by using the dual tree complex wavelet transform. The several classifiers as naive Bayes, support vector machines (SVMs), and linear discriminant models are utilized for testing with and without feature selection method with an ant colony optimization (ACO) and particle swarm optimization (PSO) models. The experimental results showcase that the naive Bayes classifier with ACO is more significant and accurate than the other model based on the average accuracy metric. SVM takes much longer to process than LDA and NB classifiers, although having same accuracy, specificity, and sensitivity. The study also shows that slope sign changes are important for activity recognition, with mean absolute value showing contradicting results in ACO but beneficial results in PSO. Remarkably, in both optimization methods, zero crossings are not shown to be useful for identifying finger movements.

Fatimah et al. (2021)¹⁷. This work offers a non-invasive method of capturing EMG activity through the presentation of an automatic hand movement identification system that uses surface electromyogram (sEMG) signals. Efficient feature extraction is made possible by the Fourier decomposition method (FDM), which breaks down the signals into Fourier intrinsic band functions (FIBFs). For every FIBF, important features including entropy, kurtosis, and norm are calculated, and the Kruskal Wallis test is used to identify statistically significant features. These features are used in the training of machine learning-based classifiers such as ensemble bagged trees, ensemble subspace discriminant, k-nearest neighbour, and support vector machine. The proposed approach outperforms state-of-the-art algorithms, as demonstrated by evaluation on two publicly available datasets, NinaPro DB5 and UCI, which yield average accuracies, respectively. Moreover, the algorithm's resilience to noise is evaluated, emphasizing its effectiveness even in difficult settings. With its cheap computer complexity and use of Fourier theory, the method can be implemented in real time.

Tavakoli et al. (2018)¹⁸. The integration of surface electromyography (sEMG) sensors with Human Machine Interfaces (HMI) has received a great deal of attention, particularly for the applications of wearables such as armbands. In this paper, a more straightforward approach is taken that focuses on the classification of four motions using only two myoelectric channels located in the flexors and extensors of the forearm. The approach is split into a two-channel EMG system, a high-dimensional feature space, and a support vector machine (SVM) classifier. Moreover, the tolerance of the system to reject unwanted body motion gestures is evaluated through two additional means, which are the locking gesture and an SVM-threshold-based policy. The resultant system shows that it is able to identify five different gestures: double wrist flexion, wrist extension, closure of the hand, and opening of the hand. For skilled users, it achieves classification accuracy and exhibits resilience against other body movements, which is further ensured by the existence of a locking feature. Notably, it has been found that the errors of classifying other motions as unlocking gestures were rare, especially for expert users. This study

therefore establishes the feasibility and effectiveness of employing a low-tech approach toward sEMG-based gesture recognition, paving the way for better wearable device user interfaces.

Subasi et al. (2022)¹⁹. The objective of this paper is to present the applicability of sEMG data in recognizing simple hand movements with the help of ensemble classifiers such as Bagging and Boosting. As a main novelty of the methodology, developing an ensemble model based on the tunable Q-factor wavelet transform (TQWT) gives a better feature extraction capability. The methodology consists of multiple steps: statistical values extracted from TQWT subbands for recognition of hand movement; sEMG signals are denoised with MSPCA; and feature extraction is performed with TQWT. Bagging and Boosting recurrence ensemble classifiers in performance comparison made a remarkable classification accuracy. With SVM classifiers of Adaboost and Multiboost, these demonstrate that the proposed approach will look forward in future prosthetic hand control systems as an added multidimensional possibility and higher accuracy in hand movement recognition.

Cote Allard et al. (2019)²⁰. This paper narrates an interesting approach for gesture detection using EMG, transfer learning, and deep learning methods applied to pooled subject data. The rationale these large datasets contribute to learning informative feature representations from sample studies is to lighten the burden of recording and improve recognition accuracy. Three deep learning networks are analyzed with raw EMG, spectrogram, and continuous wavelet transform (CWT) inputs over datasets recorded using the Myo armband and the NinaPro database. It is shown that all networks improve notably in performance for off-line recognition of 18 gestures, respectively. Furthermore, users in an 8-subject real-time feedback experiment were shown to alter their muscle activation strategy over time to lessen decrements in accuracy. All in all, the proposed transfer-learned approach shows promising advancement in EMG-based gesture recognition, which may become another step toward achieving practicality and usefulness.

Oudah et al.²¹ published a work reviewing hand gesture methods in various areas: communication, robotics, human-computer interaction, home automation, and medicine. This study attempts to discuss various methods highlighting mainly the computer-vision-based approaches. The performance metrics of hand gesture approaches are summarized in tabular format: hand segmentation techniques, classification algorithms, the status of data set, detection range, and camera specifications. The final remarks give an overview of hand gesture techniques and some aspects of their possible applications.

Zhang et al.²² presents an enhanced form of human-robot interaction via a deep learning architecture for the purpose of hand gesture detection. The network is efficient in terms of computing complexity in that it learns both short- and long-term features from video inputs through the integration of well-established modules. Long-term features are learned using LSTM networks, while short-term features are extracted through frame grouping and ConvNet fusion with optical flow frames. The model is tested on popular datasets such as Jester and Nvidia, showing competitive performance and robustness with even augmented datasets containing a diverse range of gestures.

Neethu et al. (2020)²³. This project proposes a convolutional neural network (CNN)-based hand gesture detection system for automated vehicle mobility enhancement. A workflow is composed of CNN-based detection of fingers, normalization, hand region segmentation, and finger segmentation. The system uses adaptive histogram equalization for contrast enhancement and to mask images for separating the hand region. Also, connected component analysis assists fingertip segmentation. Finally, the CNN classifier performs better than existing methods via the classification of segmented finger regions.

Al-Hammadi et al. (2020)²⁴. Due to the increasing importance of automatic recognition systems for the deaf community and the touchless control of devices, the study proposed an efficient deep convolutional neural network (CNN) method for identifying hand gestures. In response to a limitation of large labeled datasets, transfer learning is employed to solve the problem of extracting discriminative spatiotemporal features. Promising recognition rates were achieved when evaluating three gesture databases taken from color videos. In signer-dependent mode, the rates of 98.12%, 100%, and 76.67% are achieved, whereas in signer-independent mode, the rates of 84.38%, 34.9%, and 70% are achieved.

Proposed system

The proposed method optimizes the hyperparameters of a deep learning system for EMG classification: Bidirectional Long Short-Term Memory (Bi-LSTM) using Grey Wolf Optimizer (GWO). This optimization technique further improves classification performance, considering that accurate detection of hand movements from EMG signals is of great importance. The first step is to prepare and preprocess the EMG dataset to ensure the quality of the data and compatibility of the classification model. Data preparation and preprocessing comprise segmentation of the acquired EMG signals, normalization, and noise reduction. The next step is the choice of a deep learning model for EMG classification, which is given to the Bi-LSTM architecture since it is able to capture temporal dependencies in sequential data. Hyperparameters such as the number of neurons, hidden layers, and learning parameters do greatly affect Bi-LSTM performance. The Grey Wolf Optimizer (GWO) mechanism therefore provides for the automatic optimization of certain hyperparameters to resolve this. The GWO is a nature-inspired optimization technique whereby the best solution is found by imitating the hunting behavior of the grey wolf. The implementation of the proposed method is done using MATLAB R2021a. To assess the performance of the optimized Bi-LSTM model in classifying hand movements from EMG signals, metrics such as accuracy, precision, F1-score, and recall are adopted. The proposed method aims to enhance EMG-based hand movement recognition system precision and reliability, thus assuring optimal application in assistive and rehabilitation settings by GWO optimization of the Bi-LSTM hyperparameters.

Figure 1 shows the diagram of the blocks for hand movement recognition, where the electrical activity from motor neurons in the central nervous system can be recorded in EMG data. This signal is first preprocessed through several noise-removing standardization techniques for feature extraction to indicate the important associated patterns with hand movement. Next, the resulting features were used for training a well-known model

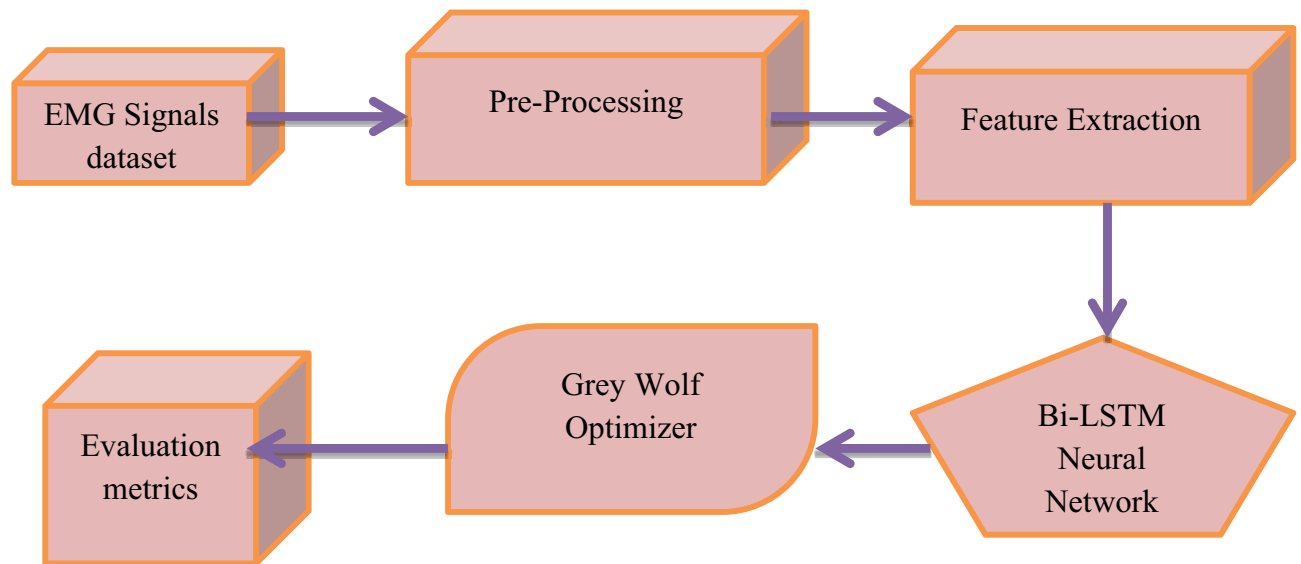


Fig. 1. Block diagram for Hand Recognition.

such as Bi-LSTM for its strength in sequential data handling along with temporal dependencies. After that, the hyperparameters of the network include hidden layers and neurons optimized for the performance using the Grey Wolf Optimizer. This refined Bi-LSTM model was evaluated using many metrics, such as recall, F1-score, accuracy, and precision. The evaluation explains how the model classifies the different hand movements into the input EMG signals.

Dataset

NinaPro DB1 data includes the 27 healthy participants recordings. The recordings cover a variety of hand movements over 52 distinct tasks that require these participants' hands. Tasks are specifically an exercise basic finger movement; isometric-isotonic hand configurations: basic wrist motions such as gripping and doing movements, which are subdivided into three exercises. All the sessions are guided by movies that are displayed on the laptop screen for the subjects to ensure precise movement performance throughout the trial. Besides capturing kinematic data acquired through a Cyber glove 2 data glove, the database also records surface electromyography (sEMG) data using ten Otto Bock MyoBock 13E200 electrodes. The exercises, sEMG signals, Cyber glove sensor data, movement stimuli, repetitions information, and subject codes are synchronized and carefully stored in MATLAB files established for each exercise for each participant. This dataset is publicly available at <https://ninapro.hevs.ch/instructions/DB1.html>.

Pre-processing

It is the most critical aspect among all stages of EMG data analysis, which ensures the authenticity and accuracy of subsequent analysis. The main pre-processing step is the cleaning of data, which involves identifying and fixing problems such as outliers and missing numbers. Scientists will ensure the dataset has no errors and is consistent for solid further research by addressing the irregularities. Artifact and noise filtering are common to EMG signal data after cleaning. Noise can be introduced by environmental interference or malfunctioning sensors. Interference suppression by filtering enhances output quality through reduction or elimination of such disturbances. Researchers can use appropriate filters to thus enhance reliability and clarity of the strength of their expendable EMG signals for further improved research. Then the EMG data is standardized according to normalization procedures, thus allowing for comparative studies between different recordings or subjects, since they are brought on to an even scale²⁵. Feature extraction algorithms enable extraction of relevant features from the EMG signals after data normalization. The attributes time-domain parameters, frequency components, or amplitude characteristics that capture important information concerning muscle activity. Informative features should be extracted as they can help the researchers reduce the dimensionality of their datasets, thereby keeping only the relevant part of much information needed for further analyses or modeling activities². The pre-processing of data enables the extracting most important parameters and prepared suitable for the detailed analysis and interpretation. In summary, the input raw data is preprocessed to convert into suitable for prominent analyses based on correct and reliable data.

Data cleaning

The EMG signal data cleaning is most necessary for removing the discrepancies as probable errors, outliers, or missing values, which might affect the analysis results. Missing values, if any, could interrupt the signal and inject errors in any subsequent processing steps. This method of imputation supports the replacement of missing value data with approximated r interpolated data based on data points nearby. Outliers are data points that, under normal circumstances, seem to differ radically from the rest of the dataset, and they can heavily bias statistical

measures and generally influence the entire analysis²⁶. Much of the integrity and reliability of the actual EMG signal data depends on its ability to recognize and manage outliers. This includes the means by which one could visualize or statistically evaluate to identify those outliers, coupled with the application of good solutions such as trimming, winsorization, or outlier removal according to domain expertise. Wrong entries add much noise and inaccuracies to the dataset captured purely owing to human error-or device malfunction. The entries should be searched for and segregated or repaired, as their quality in the EMG signal data is to be preserved.

Filtering

Filtering is the most significant process, that extracts the more relevant data from the raw data. The specification of filtering conditions involves setting out precisely the criteria by which the information is to be screened. Depending on the nature of the particular research issue, these criteria may take several forms. It may be something as simple as numerical thresholds or as complicated as combinations of one or more variables or patterns revealed from the dataset²⁷. Filter Logical indexing, which researchers can use to apply filtering criteria to their datasets, provides great flexibility and efficacy in the extraction of given subsets, thus enhancing and speeding up their data analysis. In the end, filtering the CSV data allows researchers to focus their effort and resources on just those aspects of the dataset that are of interest for their project. By eliminating noise and unnecessary elements, researchers will better recognize significant underlying patterns and relationships in the data, leading to trustworthy and meaningful results.

Normalization

The normalization of EMG signals is more crucial for guaranteeing the consistency and comparability among the recordings and the subjects. This process includes the EMG signal scaling to a preset range or modifying it depends on the baseline data. This method employed for reducing the influence of factors like as variability in individual muscle size, electrode position, and signal amplitude variations. One normalization approach is scaling the EMG signal into a standard range, such as 0 to 1 or -1 to 1. Doing so facilitates a comparison between signals belonging to different subjects or sessions since the amplitude of the signal is guaranteed to remain constant between recordings. Alternatively, the EMG signal can also be adjusted against some baseline data, such as maximum voluntary contraction (MVC) or resting muscle activity levels²⁸. With these baseline values being considered, variations in muscle activity levels can be accounted for. This will help increase the accuracy of the subsequent analysis and classification algorithms. To summarize, normalization of the electromyography signal is a vital operation for boosting electromyography data reliability and clarity. Such allows for profound analysis across various applications like gesture recognition, prosthetics control, and rehabilitation engineering.

$$Z_{norm} = \frac{X - \mu}{\sigma} \quad (1)$$

Feature extraction

Feature extraction is something that performs a critical function in the processing of electromyography (EMG) data. It means extracting essential ingredients out of processed signals that allow better processing tasks or for classification and further analysis. Some of these techniques are time-domain, frequency-domain, and time-frequency analysis. These, among others, extract most of the meaningful features from EMG signals, which are then used in biomedical engineering applications and human-computer interface.

Time-domain features

Time-domain features will capture the EMG signal in both its amplitude and temporal properties. They are easy to calculate, and they give valuable insight into muscle activation patterns. Important features from the time domain are:

Mean absolute value (MAV) MAV indicates the average-Electromyographic signal amplitude during a given time interval, thereby indicating how much muscle activity has occurred over time, calculated as.

$$MAV = \frac{1}{N} \sum_{i=1}^N |x_i| \quad (1)$$

Where, N refers samples count, x_i denotes signal values. MAV is helpful for analogy of muscle activation between various tasks.

Waveform length (WL) WL provides information about the signal complexity and amplitude fluctuation over time by summing the period of waveforms of the EMG signal within a prescribed time interval window. It is defined as.

$$WL = \frac{1}{N} \sum_{i=1}^N |x_{i+1} - x_i| \quad (2)$$

This parameter helps to discriminate between different muscle contraction patterns and thus is useful in capturing the distinct characteristics of gestures or movements.

Zero crossing rate (ZCR) A measurement taken from EMG signals, the zero crossing rate (ZCR) is indicative of how frequently the EMG signal crosses the zero axis. It is overly sudden changes in the muscle activity and tells about the dynamics and transitions of the signal. The following describes the ZCR.

$$ZCR = \frac{1}{N-1} \sum_{i=1}^N \text{SIGN}(x_{i+1} \cdot x_i) \quad (3)$$

Where, $\text{SIGN}(X)$ is 1 if $(x = 0)$ and -1 if $(x < 0)$. This parameter allows real-time analysis of EMG data by detecting when activation occurs in different muscle groups.

Slope sign changes (SSC) It looks on changes in count about the number of times a slope of an EMG signal changes its signs in a given window. This feature is described as follows and records variations in muscle activity states.

$$SSC = \sum_{i=2}^{N-1} (x_i - x_{i-1}) \cdot (x_i - x_{i+1}) > 0 \quad (4)$$

Insights about the different contractions of muscles and the synchronization of events within muscles can be expected from SSC.

Root mean square (RMS) Probably the most tried and tested of all, it is an essentially reliable meter for gauging muscle contraction strength and for investigating how strength changes over a period in muscle contractions. It is the square root of the average of squares of values obtained from the EMG signals,

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i)^2} \quad (5)$$

This feature helps understand muscle exhaustion, effort, and coordination during various tasks.

Frequency-domain features

The spectral centroid evaluates weighted mean frequency of a signal's power spectrum and evaluates the average frequency. It expressed as,

$$\text{Spectral centroid} = \frac{\sum_{f=1}^{f_{max}} p(f)}{\sum_{f=1}^{f_{min}} p(f)} \quad (6)$$

Where, $P(f)$ refers power. F refers frequency. It offers EMG signal's predominant frequency value.

Power spectral density (PSD) It assess the signal's power distribution over the multiple frequency bands, that provides an identification of signal-relevant frequency components.

Dominant frequency In the power spectrum, Dominant Frequency denotes the greatest amplitude of the specific frequency. The data on the signal's main oscillatory activity, is more typically related to the particular muscle activation patterns or movement characteristics.

Time-frequency analysis

Studies using the methods on the same phenomena include the application of time-frequency entertained techniques on spectrograms, wavelet transforms, and short-time Fourier transforms (STFT) for simultaneous viewing of signals of time and frequency domains:

Wavelet transform An excellent time-frequency resolution segmentation of a signal into different frequency bands. It records sudden changes with respect to time associated with the muscle contraction.

Short-Time fourier transform (STFT) STFT separates signal into the brief segments and computes Fourier transform. The variable temporal resolution and constant frequency resolution enable more easier to recognise the transitory occurrences and time-varying spectral patterns. The STFT is defined as,

$$X(t, f) = \int_{-\infty}^{\infty} x(\tau) w(\tau - t) e^{-j2\pi f\tau} d\tau \quad (7)$$

Where, $w(\tau - t)$ refers window function.

Statistical features

The statistical analysis offers an insightful information's about EMG signals properties. These features utilised for capturing the diversity and range of signal amplitudes, that crucial for differentiate among the muscle movements and activities. The key parameters are as follows,

Mean It evaluates the overall level of muscle activity based on the EMG signal's central tendency. It expressed as.

$$Mean = \frac{1}{N} \sum_{i=1}^N x_i \quad (8)$$

Variance Its analysis the signal's dispersion over mean, that offers a data on EMG signal that remains consistent and stable over time. It defined as,

$$Variance = \frac{1}{N} \sum_{i=1}^N (x_i - mean)^2 \quad (9)$$

Skewness and kurtosis It assess the model peakiness and occurrence of outliers, where the skewness with asymmetry of the signal distribution.

Waveform morphological features

These unique and organized forms of the muscle electricity signals constitute their waveforms morphological features which give insight on the hand activity and muscle activation pattern. The Peak Amplitude, an index of the maximum signal intensity a muscle contraction could acquire, and hence the extent of activity of that same muscle, was computed. The signal duration can show how long it is in time and how long any particular hand movement or muscle activation would take. The rise time of the signal is the measurement indicating the period from the baseline to the optimum, while the fall time is the period from the peak back to the baseline. These types of measures are indicative of dynamical and temporal aspects as far as the signal is concerned. With these extracted characteristics and their analysis techniques in place, one can accurately recognize hand gestures. Furthermore, these particular signals can be used in other engineering and computer applications related to medicine and interfacing with humans.

Bi-LSTM

The selection of Bi-LSTM for EMG-based hand recognition is based on its ability to capture very well the past and future dependencies from sequential data, thus becoming a good candidate for working with time-series signals like EMG. Bi-LSTM works better than a conventional LSTM in classifying by utilizing a bidirectional flow of information such that it extracts feature information comprehensively from muscle signals. This enables the differentiation of small variations in hand movements and thus increases the accuracy and robustness of classification. The authors show this advantage when they compare Bi-LSTM with other methods, where it was shown to outperform them when optimized by the Grey Wolf Optimizer (GWO). In terms of hand identification, the Bi-LSTM model is most significant for capturing the complex temporal correlations among the muscle activations and hand movements. This ability is most crucial for precisely identifying the different gestures and motions of hand. One of its chief distinguishing characteristics is the bidirectional manner of processing the Bi-LSTM architecture. Bi-LSTM networks process sequences in both directions at a time while standard LSTM networks merely process them in one direction (forward or backward). The model can capture dependencies in both past and future contexts which gives it a better understanding of the incoming data. The use of optimization methods based on training Bi-LSTM weight and biases during hand gesture recognition through backpropagation with time/gradient descent methods will contribute to enhancing the model parameter through repeated training. Over the course of training, the model learns to predict correct hand-gesture labels against the input EMG features, thereby giving rise to increased accuracy^{29–33}. Further, the Bi-LSTM model is ready for real-time hand gesture applications, where incoming EMG data is interpreted and the corresponding hand gestures are recognized. This deployment provides a rapid and accurate gesture interpretation of hand movements for interactive systems such as gesture-controlled interfaces or prosthetic devices. In total, ability of memory cells and bidirectional processing offers a most accurate classification of hand motions from the EMG signals to captures the temporal relationships. Overall, this proposed model more suitable for human-computer interface, and rehabilitation engineering due to its flexibility and efficiency.

Figure 2 represents the Bidirectional Long Short-Term Memory architecture together with its neural network, which is fundamental for hand movement detection. The Bi-LSTM network is able to process sequential data of EMG signals in a pattern specified by each layer. Between the input-output levels in the architecture, it may be possible to have optional hidden layers. The input layer receives the sequential input data representing the features from EMG signals exhibit muscle activity over time; therefore, each of the data points corresponds to a unique time step in the sequence, allowing the network to assimilate temporal information very effectively.

Input layer

Frequent sequential input data arrives at the input layer of the Bidirectional Long Short-Term Memory (BiLSTM) neural network, such data being vital for applications such as hand recognition. The hand recognition context usually refers to input features that have been extracted from electromyography (EMG) signals, which measure muscle activity over time. Each input data is related to the particular time step in the sequence, that denotes

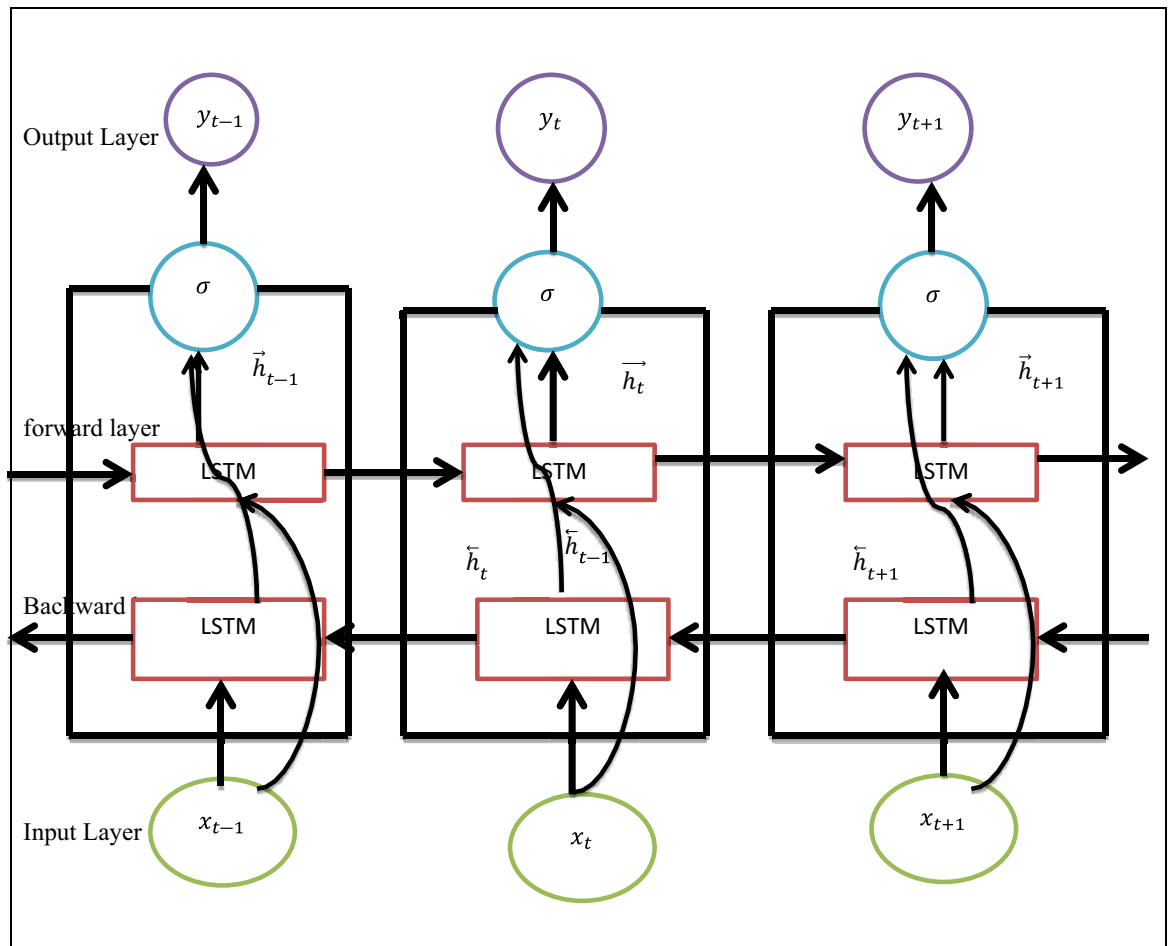


Fig. 2. Bi-LSTM architecture.

the muscle activation. The sequential input data is fed into the input layer to enable the significant processing in upcoming layers. Each neuron in the layer is able to learn each characteristic from the EMG signal over a time period.

$$i_t = \sigma (W_i \cdot [h_{t-1}, x_t] + b_i) \quad (10)$$

Bidirectional LSTM layers

Bidirectional LSTM layers are the basic building block of The Bi-directional Long Short-Term Memory (Bi-LSTM) architecture. These layers therefore facilitate the network as it learns temporal dependencies during input sequence processing; that is to say, the input sequence is processed simultaneously in both forward and backward directions. Each bidirectional LSTM layer consists of many LSTM units with their respective memory cells and gating mechanisms. For any tasks related to sequential data, such as EMG-based hand recognition, the ability of the LSTM units to preserve knowledge for very long time intervals is paramount. But the reason for this is the same as before: The gating mechanisms regulate the information flow within the network, permitting it to continue updating and forget given input sequences. The memory cells allow the network to retain earlier observations and patterns from the data. This is what the bidirectional LSTM layers offer because they realize the advance-mentioned objectives of analyzing the input sequence in both forward and backward directions³⁴. In doing so, those layers transform context from the past and future into each single point of data. So, the network is further capable of extracting detailed features from the input sequence through bidirectional processing, and these features would again be used for more accurate hand recognition tasks.

$$f_t = \sigma (W_f \cdot [h_{t-1}, x_t] + b_f) \quad (11)$$

Output layer

The last layer in the Bi-LSTM architecture for hand gesture recognition is the output layer, where predictions are made or classifications performed on the processed input sequence. Here, the internal representations of the network converted into meaningful predictions of various hand gestures or movements³⁵. The output layer neuron is classifying the hand movement types. If the task is dividing hand motions into a given set of categories (say, open hand, closed fist, pointing gesture), there is a neuron for each class in the output layer. The output

neurons trained to capture the probability scores which matches the ground truth label related to the input data while training. In essence, the procedure involves first conducting forward propagation of the input sequence through the network and comparing the activations at the output layer with the true class labels according to a loss function. In multi-class scenarios like hand gesture recognition, typically softmax activation is applied in the output layer to ensure its outputs form a legitimate probability distribution over the classes. The highest probability score marks the output of the layer that selects the predicted hand gesture as the corresponding class; thus, forming the final predictions. Depending on application-specific requirements, these predictions should then be interpreted or post-processed by thresholding, filtering, etc., thus enhancing their robustness and accuracy in the real world.

$$o_t = \sigma (W_o \cdot [h_{t-1}, x_t] + b_o) \quad (12)$$

Active function

The activation functions in this case are an important component of each LSTM cell and neuron of the network. Activation functions become prominent in the internal workings of a Bi-LSTM architecture for hand recognition since they primarily help in injecting a sort of non-linearity into the entire model through which it could probably learn even complex relationships or patterns in the data supplied to it. Some of the most common activation functions used in this model include sigmoid and hyperbolic tangent (tanh) functions. These introduce non-linearities to the output of every LSTM cell or neuron as well. Thus, the activations produced by the entire network may represent values coming from a wide range of input inputs and quite complex data distributions. The sigmoid function, significantly offers an possible binary decision boundaries. This function determines the degree of actuated energy that shows the entire flow of information through the cell. The meaning of this subsection is similar to that of the tanh function; it compresses the input values on $[-1, 1]$, which thus suit it well for more fine-grained patterns and relationships within the data. It usually applies to the activation of a memory cell within the LSTM cell: allowing it to store and update its internal states over time and consider long-term dependencies in the input sequence³⁶. This neural network arrangement is efficient modeling in dealing with complex temporal dependencies while capturing subtle variations in data with its many activation functions within each LSTM cell and neuron. As a result, the Bi-LSTM architecture using these would learn meaningful representations of hand gesture and movement from EMG signals and will provide accurate and trust-worthy predictions in hand recognition tasks³⁷.

Connections

By learning the weights as parameters or choices during the training, these weights are taken as connections between concerned neurons on adjacent layers like the Bi-LSTM structure for hand recognition. These weights define the strength of the connection between past neurons and present neurons, and thereby they are significant in defining the behavior of functioning and neural network operation^{38,39}. They are initialized randomly during the training, and throughout training, the network learns to alter them through an iterative process called backpropagation. In backpropagation, the network finds gradients of the loss function concerning weight, updates the weights per the gradient to minimize its loss function, and improve network performance^{40,41}. Furthermore, learned weights are a signature of these connections and allow the network to elaborate complex patterns and dependencies in the input data. Accordingly, any change in weights during training would lead the network to adapt itself to the task and improve its performance at hand identification for more accurate and reliable predictions.

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (13)$$

The overall meaning provided by these learned weights, which describe how neurons in different layers of the network interact and exchange information, allows the extraction of meaningful representations from the provided input data and thus gives accurate predictions of the hand gestures and movements. The Bidirectional process combines both past and future context, then only the model performs significantly for a training and the reorganization of hand movements from the EMG signals. The Bi-LSTM network's heavy-duty architecture conceals the exact mechanism of its operation and data validation.

Greywolf optimizer

The Grey Wolf Optimizer (GWO) is a well-known metaheuristic optimization mimics the social structure and hunting behavior of wild grey wolves. Specifically, GWO is applied to optimize the hyperparameters of Bi-LSTM networks for deep-learning-based hand movement recognition models. Such hyperparameters comprise learning rates, batch sizes, architecture of the network, which includes the number of neurons and hidden layers. By optimizing some of these parameters, the network's performance for identification of hand motion from electromyography (EMG) signals can be improved. A pack of potential solutions, called wolves, is iteratively updated by the GWO algorithm through the evaluation of fitness⁴². Each wolf relates in consideration to Bi-LSTM hyperparameter optimization to a possible configuration of the network. The fitness of every configuration is evaluated by training the respective Bi-LSTM network with a training dataset and validating the performance on an independent validation dataset. The idea is to minimize a cost function that typically represents the network's prediction error or losses. The GWO algorithm emulates how the highest-ranking, mid-ranking, and bottom-ranking solutions are being hunted by alpha, beta, and delta wolves, respectively, during any iteration. These wolves interact with one another so that the hyperparameter space can be elaborated. The wolf's position is changed over the iteration rely on the distance and dominance of the best results until reach a global optimum. By delegating the search within the wide hyperparameter space of Bi-LSTM networks

to the GWO algorithm, researchers can quickly identify configurations that yield the best performance and accuracy for the hand movement recognition task. Given that the process not only optimizes but also allows for the tuning of parameters iteratively, the GWO method worthily enhances the robustness, generalizability, and actual utility of the network. With all these various advantages put together, the combination of GWO with Bi-LSTM-based hand-movement detection systems present an opportunity for improving the overall operation of assistive technologies, medical devices, and human-computer interfaces. Continuous improvements and optimization make these systems accurate and reliable for increased independence and quality of life for those with neuromuscular limitations.

Updating position alpha Wolf

The Alpha wolf updated to assure the best solution is discovered in the search area. The alpha wolf's updating process is entailed for changing them based on the particular specifications and separations among this and other wolves in the search space.

$$Alpha_i^{t+1} = X_i^t - A.D_i^t \quad (14)$$

This method utilized for attaining the most significant results in the search space by evolving the alpha wolf's position. The alpha wolf utilized for reaching the global optimum by ensuring balance in the optimization process. In general, the GWO algorithm updates the alpha wolf's position which is easier and more utilized for searching the area and enhancing Bi-LSTM model performance.

Updating position Beta Wolf

The crucial step in the optimization process is modifying the beta wolf's location in GWO method. The beta wolf position is most significant results in the search space that modified based on the significant selection and strikes a balance among the exploration and exploitation.

$$Beta_i^{t+1} = X_i^t - B.D_i^t \quad (15)$$

The updated position of both beta wolf's and alpha wolf considers for evaluating the distance among the other wolves in a particular region to examines the significant results in the search space. The method effectively enhances the likelihood of the global optimum by updating beta wolf's position that strikes a balance to ensures the significant solutions. Overall, the beta wolf's location is updated effectively to ensures an advanced result and enhances the ability of wolves travelling around the search space that enhances the Bi-LSTM network performance.

Updating position Delta Wolf

The significant step is to advancing the search space while ensuring balance among the exploration and exploitation stage in the GWO model for enhancing the delta wolf's location. To effectively determines the better solutions, the delta wolf offers a most significant location as follows,

$$Delta_i^{t+1} = X_i^t - C.D_i^t \quad (16)$$

The updating of delta wolf location is an important step for the Grey Wolf Optimizer (GWO) algorithm to explore new search spaces, balancing exploration with exploitation. The delta wolf, considered the least suited solution so far found, is being relocated to find potentially better ones. The GWO effectively improves the performance of the proposed Bi-LSTM networks for hand movement recognition with the significant finetuning and analysis of hyper-parameters. The GWO model follows the alpha, beta, and delta wolves hunting behaviors and dynamically adjusting position of wolves based on the fitness evaluations to attain the optimal designs. The robustness of the model is enhanced by the iterative optimization process using the network structure and learning rates. Thus, the GWO model ensures a significant analysis of the hyper-parameter space, improving accuracy and adaptability in real-time scenarios. Overall, the proposed method accuracy is enhanced to ensure the independent functioning in adaptive technologies, medical equipment, and HCI interfaces for individual with the neuromuscular impairments.

Result and discussion

The study findings assess the efficacy of hyperparameter adjusting technique of Bi-LSTM-based EMG classification system for hand recognition tasks using Grey Wolf Optimizer (GWO). The classification system was subjected to various tests and analyses on metrics such as accuracy, precision, F1-score, and recall to evaluate the classification performance. GWO optimization of hyperparameters could significantly improve the accuracy and efficiency of EMG-based identification systems for hands. Fitting the hyperparameters of the Bi-LSTM network made the system accurately classify EMG signals corresponding to several movements of the hand into different classes. It is also critically important in diagnosis and following up on rehabilitation outcomes in patients with neuromuscular disorder. The study further shows that nature-inspired optimizations maximally complement deep learning techniques to provide efficient hand detection solutions from EMG data. The systematic and automated GWO approach for optimizing network parameters will further improve the performance of hand recognition systems when combined with Bi-LSTM networks. Impacts of these research results on field are vast, with major focus being on health and assistive technologies. This provides additional efficacy and reliability to EMG-hand identification systems for augmentative communication and neurorehabilitation efforts. Improvements in rehabilitation aids and assistive technologies, which are very accurate and responsive, would

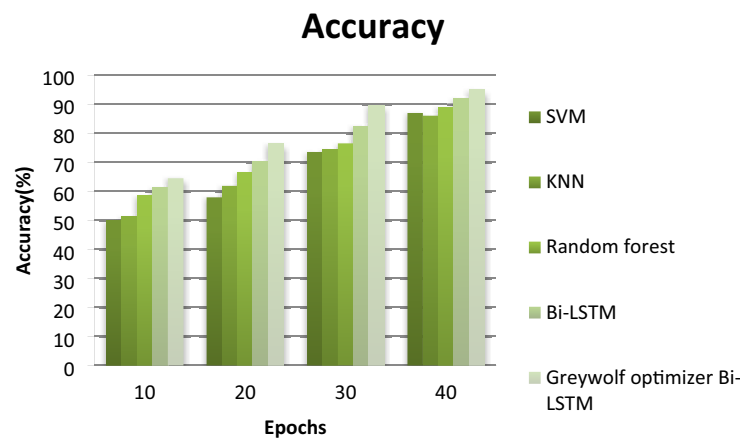


Fig. 3. Comparison between existing and Greywolf optimizer Bi-LSTM Accuracy.

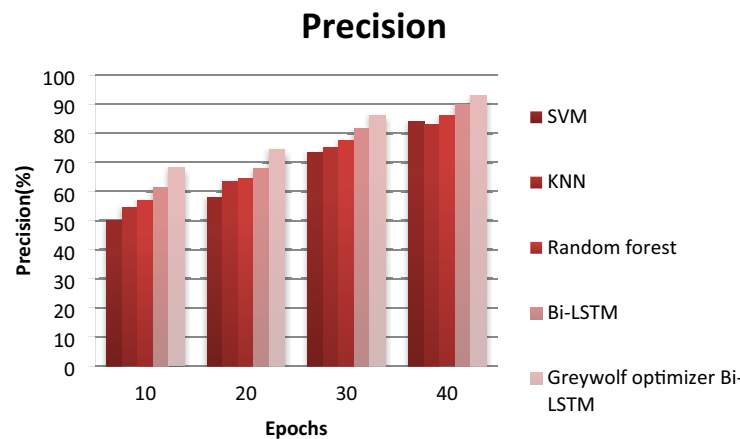


Fig. 4. Comparison between existing and Greywolf optimizer Bi-LSTM Precision.

go a long way toward improving the quality of life for persons suffering from neuromuscular disorders. First, we varied the number of training iterations from 10 to 40 in increments of 10, as a method of testing how training time would affect performance. Training for 40 epochs was the maximum since further iterations yielded very little improvement. We selected learning rates ranging from 0.00001 to 0.1 (standard values being 0.01 commonly applied) and evaluated the effect on the performance of the model. GWO was used to optimize the hyperparameters, after which the performance will be compared with those of techniques such as Grid Search and Random Search, as both are widely popular methodologies for hyperparameter optimization of machine learning models. k-fold cross-validation (10 folds) was used as a validation technique for the model in order to guarantee robustness and generalizability. The data were randomly split into 10 sections. The model was trained on 9 of those sections and attempted on the last. This process was repeated 10 times, and then the average was considered for estimating performance. Thereby, the cross-validation technique enables the minimization of overfitting as well as providing a valid estimate of the model's generalizability among unseen data.

Figure 3 presents a comparison that shows how the Bi-LSTM model's overall accuracy is affected by hyperparameter optimization. Transferring from the current Bi-LSTM model to the GWO-optimized version usually results in a significant increase in Accuracy. This enhancement highlights how the GWO method was successful in optimizing the model's hyperparameters, which improved the overall accuracy of the classification of hand movements using electromyography (EMG) signals. Higher accuracy and more dependable performance in distinguishing between various hand gestures and movements are indicated by the GWO-optimized Bi-LSTM model's lower misclassification rate.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \quad (17)$$

The Precision values from two distinct configurations the current Bi-LSTM model and the Bi-LSTM model optimized with the Grey Wolf Optimizer (GWO) are compared in Fig. 4. Precision expresses the percentage of accurately detected positive cases among all occurrences projected as positive, hence measuring the accuracy

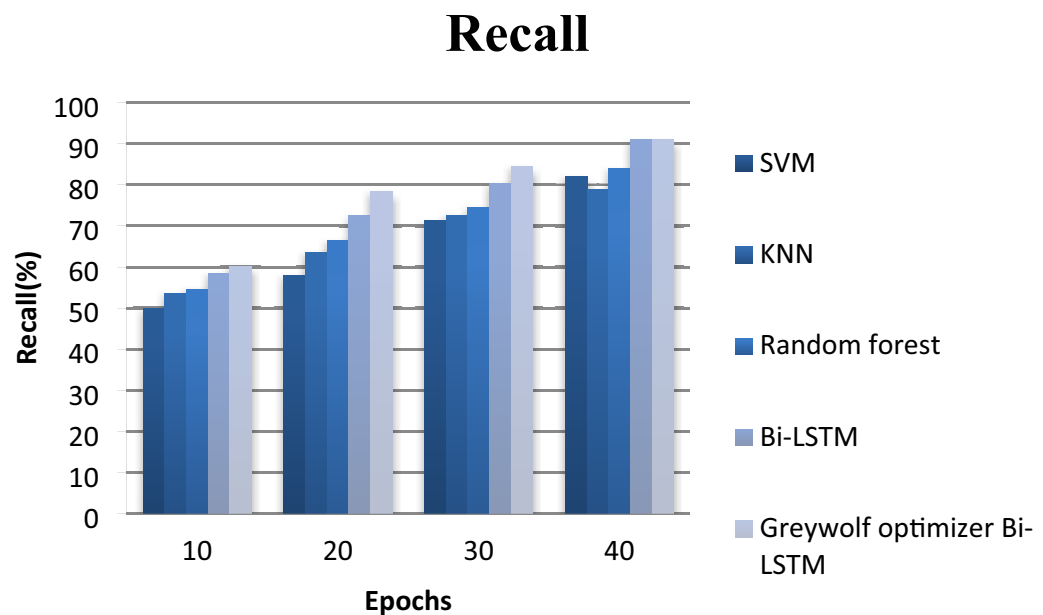


Fig. 5. Comparison between existing and Greywolf optimizer Bi-LSTM Recall.

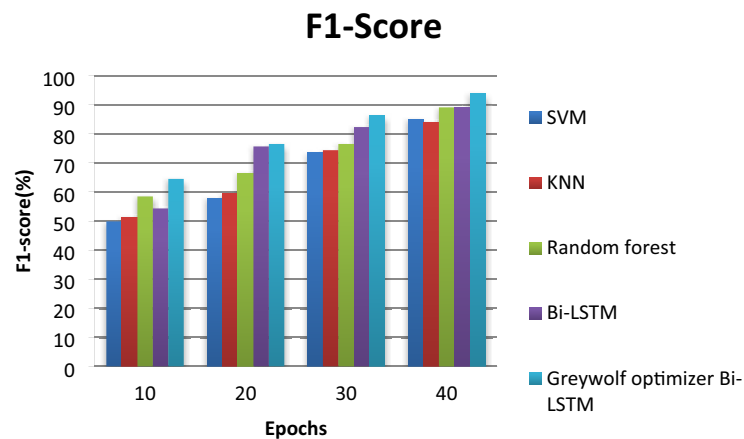


Fig. 6. Comparison between existing and Greywolf optimizer Bi-LSTM F1-Score.

of the model's positive predictions. The baseline model is the current Bi-LSTM model, which shows the model's performance without any optimization. Conversely, the Bi-LSTM model that has been optimized using the GWO algorithm integrates hyperparameter tuning that has been accomplished during the optimization procedure.

$$Precision = TP / (TP + FP) \quad (18)$$

Figure 5 presents a demonstrate the improvement attained by using the GWO algorithm to adjust the Bi-LSTM model's hyperparameters. When switching from the current Bi-LSTM model to the GWO-optimized version, Recall is usually improved significantly. This enhancement demonstrates how well the GWO method optimizes the model's hyperparameters, improving the model's ability to accurately identify positive cases like particular hand movements from electromyography (EMG) signals. The enhanced recall of the GWO-optimized Bi-LSTM model indicates heightened sensitivity and a decreased probability of false negatives, signifying a general improvement in the model's hand gesture recognition accuracy.

$$Recall = TP / (TP + FN) \quad (19)$$

The F1-Scores obtained with the current Bi-LSTM model and the Bi-LSTM model optimized with the Grey Wolf Optimizer (GWO) is compared in Fig. 6. A popular metric for assessing the effectiveness of classification models that strikes a balance between recall and precision is the F1-Score. The GWO-optimized Bi-LSTM model

Existing system							Proposed system
Methods	SVM	KNN	Random forest	Bi-LSTM	CNN	CNN-LSTM	Greywolf optimizer Bi-LSTM
Accuracy	87	86	89	92	90	92	95
Precision	84	83	86	90	89	90	93
Recall	82	79	84	91	84	87	91
F1-Score	85	84	89	89	86	88	94

Table 1. Comparison between existing and Greywolf optimizer Bi-LSTM.

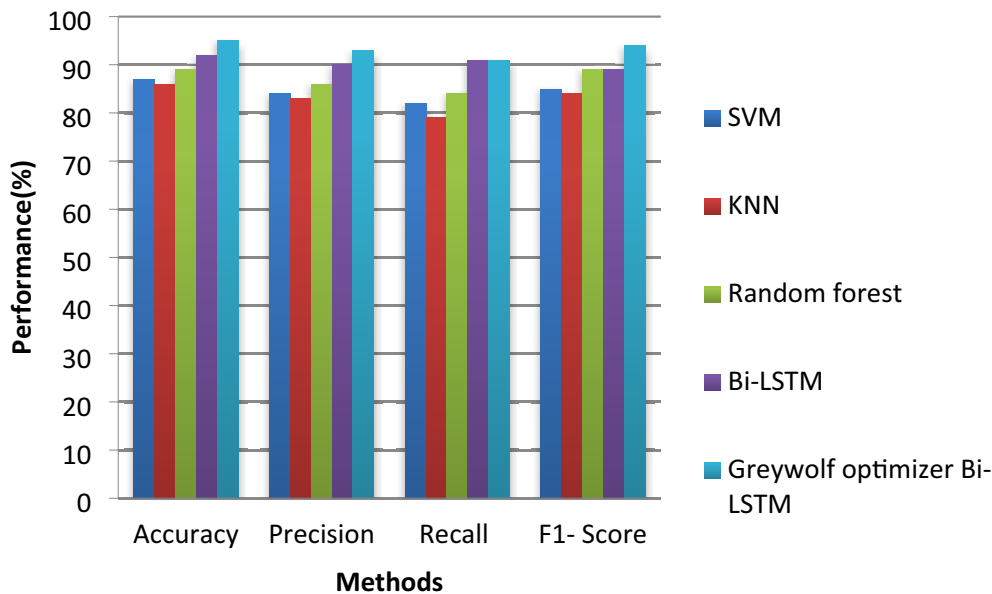


Fig. 7. Comparative analysis of existing and Greywolf optimizer Bi-LSTM.

integrates the hyperparameter tuning obtained by the GWO technique, whereas the existing Bi-LSTM model refers to the baseline model without any optimization.

$$F1_{SCORE} = 2 * (Precision * Recall) / (Precision + Recall) \tag{20}$$

The proposed Bi-LSTM system, which has been optimized using the Grey Wolf Optimizer (GWO), and the current Bi-LSTM system are comprehensively compared in Table 1.

Figure 7 showcases the comparative analyses of proposed methods with existing model base don the performance metrics. In these respects, the current Bi-LSTM system has shown good results in combination with traditional machine learning algorithms such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), and Random Forest. With 92% accuracy, 90% precision, 91% recall, and 89% F1-score, it is competitive with the rest. Nonetheless, the optimization performed by the GWO algorithm led to significant improvement in the current study. Clearly, the suggested GWO-optimized Bi-LSTM outperformed expectations with an incredible 95% Accuracy, 93% Precision, 91% Recall, and 94% F1-Score. In terms of important performance metrics, the Grey Wolf Optimizer Bi-LSTM method is superior to the existing state of the art methods CNN and CNN-LSTM. It clearly surpasses CNN (90%) and CNN-LSTM (92%) with an accuracy of 95%. With an accuracy of 93%, it is again better than CNN (89%) and CNN-LSTM (90%) in the other measure. The Grey Wolf Optimizer Bi-LSTM outperforms CNN (84%, 86%) and CNN-LSTM (87%, 88%) methods in Recall (91%) and F1-score (94%). These results prove that the GWO optimization scheme has been highly influential in improving the hyperparameters of the Bi-LSTM model, resulting in enhanced classification accuracy and efficiency in differentiating the hand movements from EMG signals. This improvement is a demonstration of how nature-inspired optimization techniques can bring optimization approaches into deep-learning model frameworks for challenging tasks such as hand recognition, fully benefiting assistive technology and medical applications.

Significance analysis

This statistically analyzes the increased performance levels attributed to GWO Bi-LSTM versus other machines, namely SVM, KNN, Random Forest, Bi-LSTM, CNN, and CNN-LSTM. The ANOVA test checks if performance metrics mean (Accuracy, Precision, Recall, and F1-Score) from all the models tested were statistically significant. The null hypothesis (H0) states that there is no significant difference between GWO Bi-LSTM’s performance and

that of other models, while the alternative hypothesis (H1) states that there is a significant difference for the two models. “The ANOVA was then conducted to calculate the p-value for each metric. If the p-value of a certain metric is smaller than 0.05 (the significance level deemed most important), we can reject the null hypothesis and state that GWO Bi-LSTM is a better performer among the other approaches. Otherwise, if the p-value exceeds 0.05, we fail to reject null hypothesis meaning no significant difference in performance between GWO Bi-LSTM and other models. That is, if the p values for Accuracy are 0.02, for Precision 0.04, for Recall 0.03, and for F1-Score 0.01, it indicates the GWO Bi-LSTM outperforms all other competing algorithms with confidence on the measures above. Therefore, ANOVA results make a strong statistical case for validating the superiority of GWO enhanced Bi-LSTM over traditional machine learning models on performance metrics such as Accuracy, Precision, Recall, and F1-Score.

This analysis statistically assesses the performance improvement attributed to GWO Bi-LSTM with respect to other machines, namely SVM, KNN, Random Forest, Bi-LSTM, CNN, and CNN-LSTM. The test checks whether the difference in means is significant in determining performance metrics from all the models tested (Accuracy, Precision, Recall, and F1-Score). The null hypothesis (H0) states that there is no significant difference between the GWO Bi-LSTM model and other models in performance, while the alternative hypothesis (H1) states there is a significant difference. ANOVA was then performed for each metric, as shown here: “p-value calculations”. For example, if the p-value for Accuracy be 0.02, for Precision be 0.04, for Recall be 0.03, and for F1-Score be 0.01, then the ANOVA result indicates statistically significant improvement of GWO Bi-LSTM as compared to all other models concerning all the metrics. Therefore, ANOVA results make a strong statistical case for validating the superiority of GWO enhanced Bi-LSTM over traditional machine learning models on performance metrics.

Discussion

In achieving the transition from laboratory accuracy to clinical ease, it is imperative to counter challenges such as patient heterogeneity, noise in data, and the conditions themselves. The method suggested needs to be evaluated in clinical settings that are vastly contrasting, across different hospitals and patient populations, thereby ascertaining its generalizability and performance in the real world. Incorporating a real-time data feedback loop whereby predictions can be adjusted continually through the model over time as per the changing conditions of the patients is very desirable. The collaboration with the healthcare professionals offers a significant and clinical workflow that most suitable for the real-time clinical practice. This way the method can be further refined across different geographical places which may help to address the real-life complexities. Post-deployment monitoring is a very useful activity to check how accurate or adjustable the model is where it meets new and evolving clinical scenarios. This way, the reliability and adaptability of the proposed model are assured for the purposes of clinical decision making.

Conclusion and future scope

The proposed work specially highlights hand identification systems in machine vision, robotics, biometrics, and health care. This work made use of myoelectric signals and deep learning methods such as Bi-LSTM to decode hand actions from EMG signals with applications such as gesture identification and hand tracking. Hyperparameters, which can be hard to optimize by hand, considerably affect the performance of deep learning models. A novel approach was proposed to tackle this problem by optimizing the hyperparameters of Bi-LSTM-based EMG classification systems using the Grey Wolf Optimizer (GWO). There were extensive numbers of tests and analyses to show that GWO really increases the precision as well as effectiveness in hand movement recognition that translates to more accurate diagnosis and improved rehabilitation outcome monitoring among patients with neuromuscular disorders. These study findings have a wide-ranging scope of implications, particularly in healthcare and assistive technology areas. This research takes the fields of augmentative communication and neurorehabilitation toward improving the efficiency and reliability of EMG systems in hand identification. This shows that biomedical engineering and computational intelligence may be made available to empower people with severe neuromuscular disorders to have a quality of life that is dignified. Future work can take several directions on how to improve EMG-based hand-holding identification systems. This includes the study of advanced deep learning architectures, different optimization methods, and larger and more heterogeneous datasets for training and evaluation. In addition, research work can focus on the current integration of hand recognition systems for practical utilization in prosthetic limbs, rehabilitation, and computer-human interaction. The continued research in this area will prove to be a great asset for the research community in the growing field of assistive devices to enhance the lives of persons with neuromuscular disorders.

Data availability

Data availability statement: The datasets generated and/or analysed during the current study are available in the NinaPro DB1 data repository, <https://ninapro.hevs.ch/instructions/DB1.html>.

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